Project Planning & Management

Project Title: Car Sales Analysis

GitHub

1. Project Proposal

Overview: The project analyzes a car sales dataset to understand market trends, vehicle pricing factors, and sales performance across different states.

Objectives:

- Find top selling car brands and models.
- Analyze how mileage and vehicle age affect the selling price.
- Compare the actual selling price with the market price to see profit and discounts.
- Build an easy and interactive Power BI dashboard for business decision makers.

Scope: The project covers data cleaning in Excel, building database tables, analyzing key business metrics, and creating interactive dashboard pages in Power BI.

2. Project Plan

The team will follow a simple schedule to finish all project phases on time:

Milestone	Deliverable	Duration
Phase 1: Planning	Project proposal and role distribution	Week 1
Phase 2: Data Cleaning	Clean dataset without nulls or wrong values	Week 2
Phase 3: Database & SQL	Create SQL tables and run business queries	Week 3
Phase 4: Power BI Dashboard	Design interactive dashboard pages	Week 4
Phase 5: Final Review	Testing numbers and preparing presentation	Week 5

3. Task Assignment & Roles

Tasks are divided equally among all 6 team members across the data analysis steps:

Team Member	Role	Key Responsibilities
Nourhan Mohammed Gaber	Data Collection & Auditing	Understand dataset columns and write project requirements.
Mai Hamada Saber	Data Cleaning - Text	Clean car data, Remove Duplicates, and data types.
Mohammed Eid Zaki	Data Cleaning - Numbers	Fix date formats, handle null and wrong values, and remove outliers.
Mohammed Alaa Korani	Data Modeling	Create database tables and relationships in SQL.
Hossam Hisham Abdelmoneim	Data Calculation	Write SQL queries and DAX measures for key metrics.
Ahmed Yasser Salah	Data Visualization	Design the Power BI dashboard layout, charts, and filters.

4. Risk Assessment & Mitigation Plan

Risk	Impact	Solution
Large dataset size causes slow loading	Medium	Clean data first and use SQL to summarize numbers before Power BI.
Missing values and incorrect prices	Medium	Review and filter out bad rows during the cleaning phase.
Sharing large dataset across teammates.	Medium	Uploading everything to GitHub to avoid resending.

5. Key Performance Indicators (KPIs)

Metrics to measure the success of the project:

- Clean and valid data with zero duplicate records.
- Answer all main business questions accurately using SQL and Power BI.
- Fast and simple dashboard navigation for users.
- Equal task contribution by all team members on GitHub.